

A Conversational Speech-to-Speech Pipeline

I would build a streaming, modular pipeline: microphone audio, acoustic echo cancellation, voice activity detection, incremental ASR, a dialogue manager and language model, then streaming text-to-speech. I assume a consent-based English assistant used in Singapore, with occasional code-switching and a reliable network. The initial goal is accurate task completion with natural interruption handling; the system must not silently perform consequential actions.

The client would send timestamped audio frames over an authenticated streaming connection. Echo cancellation would use the assistant's playback signal as a reference, preventing synthesized speech from being interpreted as a new user request. Voice activity detection would identify candidate speech regions, while the ASR service would emit provisional and finalized words. I would choose an ASR model with a supported streaming interface; a fast offline model alone does not establish acceptable streaming latency. Local recordings would test accents, names, background noise, and mixed-language utterances before deployment.

A turn manager would combine silence duration, partial transcripts, and dialogue context to decide when to respond. Short pauses should not automatically end a turn: users may hesitate or search for a word. I would tune endpointing on annotated conversations and measure premature interruptions separately from response delay. When new user speech interrupts playback, the client would stop the audio promptly and cancel queued synthesis and generation. Conversation history would record the portion actually played, so the assistant does not assume the user heard an unfinished answer.

The dialogue layer would maintain a compact session state and retrieve approved knowledge when needed. Tool requests would use validated schemas, authorization checks, and explicit confirmation for consequential actions. Retrieved text and transcripts would be treated as data, not trusted instructions. For uncertain names, numbers, or requests, the assistant would ask a short clarification rather than confidently continuing with an incorrect interpretation. Sensitive audio would have limited retention, encrypted transport and storage, and access controls; diagnostic logs would avoid unnecessary transcript content.

Streaming synthesis would start after a short, semantically stable phrase rather than waiting for the whole answer. Phrase boundaries reduce unnatural prosody and avoid speaking text that is subsequently revised. Small playback buffers would balance smooth audio against interruption responsiveness. I would set a provisional target below one second from detected turn completion to first audible response, then measure each component's contribution. This is an engineering target, not an assumed benchmark. Warm workers, bounded queues, and cancellation propagation would help control both typical and tail latency under load.

The modular design introduces error propagation: an ASR mistake can lead to a wrong answer and fluent but misleading speech. I would evaluate the complete conversation, not just individual models. Tests would include task success, entity accuracy, normalized WER, first-audio latency, interruption response time, perceived voice quality, and recovery after network failures. Results would be broken down by accent, noise condition, and utterance length, with human review of failures.

Direct speech-to-speech models offer an alternative that can retain richer timing and vocal information; Moshi demonstrates a speech-text approach to real-time dialogue [1]. I would benchmark such a model against the modular baseline using the same scenarios. I would initially favor the modular system because its transcripts, component boundaries, and replaceable models make errors easier to diagnose and controlled actions easier to verify. Singapore-focused speech data such as IMDA's National Speech Corpus would support domain evaluation, subject to its access and usage conditions [2].

References

1. Defossez et al. (2024). Moshi: a speech-text foundation model for real-time dialogue. <https://arxiv.org/abs/2410.00037>
2. IMDA. National Speech Corpus. <https://www.imda.gov.sg/about-imda/emerging-technologies-and-research/artificial-intelligence/national-speech-corpus>